Date: August 15, 2025

Algorithms -1-

University of Aleppo

Student Name:

Model: B

Duration: 90 Minutes

Student Number:

Select the correct answer for each question:

1. Calculate $\sum$ i
A. $\underline{n*(n+1)/3}$ **B**. $n*(n-1)/2$ **[1 points]**

2. What is the derivative of x^2
A. x **B**. $\underline{2*x}$ **[5 points]**

3. $x = 3+6$ what is the value of x?
A. 7 **B**. $\underline{9}$ **C**. 8 **D**. 10 **[1 points]**

4. Hello!
A. $\underline{1}$ **B**. 2 **C**. 3 **D**. 4 **[2 points]**

5. For any positive integer x what is the value of $\iiint x^5$?
A. $\underline{30}$ **B**. 40 **C**. 50 **D**. 60 **[1 points]**

6. Test
A. 13 **B**. $\underline{14}$ **C**. 15 **D**. 16 **E**. 17 **[3 points]**

7. Solve for x in the equation $2x+6=0$
A. -1 **B**. -2 **C**. $\underline{-3}$ **D**. 3 **[1 points]**

8. Test2
A. $\underline{1}$ **B**. 2 **C**. 3 **D**. 4 **E**. 5 **[3 points]**

9. Find the area of a rectangle of side $a = 6$
A. 34 **B**. $\underline{36}$ **C**. 40 **D**. 31 **[1 points]**

10. test4
A. $\underline{1}$ **B**. 2 **C**. 3 **D**. 4 **E**. 5 **[1 points]**

11. Calculate the derivative of $f(x) = x^2$ then find the limit to infinity of the function lim _{$x \to \infty$}$f(x)$
A. 14 **B**. 15 **C**. $\underline{16}$ **D**. 17 **E**. 17 **[1 points]**

12. Test3
A. 1 **B**. 2 **C**. $\underline{3}$ **D**. 4 **E**. 5 **[1 points]**

13. Please calculate $\int f(x)dx$
A. 13 **B**. $\underline{14}$ **C**. 15 **D**. 16 **[2 points]**